

A Comparative Study of Merge Sort, Quick Sort, and Heap Sort:

Theoretical Constants and Empirical Clock-Cycle Analysis

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Abstract

Sorting is one of the most fundamental problems in computer science, and the choice of sorting algorithm has real consequences for performance even when two algorithms share the same asymptotic complexity class. This essay examines three canonical comparison-based sorting algorithms — Merge Sort, Quick Sort, and Heap Sort — from both a theoretical and an empirical standpoint. We derive the recurrence relations and hidden constants that govern each algorithm’s running time, present complete, heavily commented C implementations, and measure their actual performance using clock-cycle (CPU tick) instrumentation via the standard `clock()` function. The results confirm the textbook $O(n \log n)$ behaviour of all three algorithms while revealing that their real-world constant factors — driven by memory access patterns, extra space usage, and instruction-level overhead — differ substantially, with Quick Sort emerging as the fastest in practice despite not having the best worst-case guarantee.

Contents

1	Introduction	2
2	Theoretical Background	2
2.1	Merge Sort	2
2.2	Quick Sort	2
2.3	Heap Sort	2
2.4	Summary Table	3
3	Implementation	3
3.1	Merge Sort Implementation	3
3.2	Quick Sort Implementation	4
3.3	Heap Sort Implementation	4
4	Experimental Setup	5
5	Results	5
6	Discussion	5
6.1	Confirming $O(n \log n)$ scaling	5
6.2	Merge Sort: fewest comparisons, no swaps, but slower	5
6.3	Quick Sort: most swaps, fewest cycles	5
6.4	Heap Sort: most comparisons and swaps, moderate speed	6
6.5	Constant factors versus asymptotic guarantees	6
7	Conclusion	6

1 Introduction

Sorting algorithms are usually introduced through their asymptotic complexity: the number of comparisons or swaps an algorithm performs as a function of the input size n , expressed using Big-O notation. This abstraction is essential for reasoning about scalability, but it deliberately hides the *constant factors* that determine actual wall-clock performance. Two algorithms that are both $O(n \log n)$ can differ by a factor of two or three in real running time, because Big-O notation discards constants and lower-order terms.

This essay has three goals:

1. To derive, from first principles, the recurrence relations and complexity bounds of Merge Sort, Quick Sort, and Heap Sort.
2. To present clear, fully commented C implementations of each algorithm, instrumented to count comparisons and swaps.
3. To measure actual execution time in CPU clock ticks using the C standard library's `clock()` function, across a range of input sizes, and to interpret the results in light of the theoretical analysis.

2 Theoretical Background

2.1 Merge Sort

Merge Sort is a classic divide-and-conquer algorithm. An array of size n is split into two halves, each half is recursively sorted, and the two sorted halves are merged using an auxiliary array. Its recurrence relation is:

$$T(n) = 2T\left(\frac{n}{2}\right) + \Theta(n) \quad (1)$$

By the Master Theorem, this resolves to $T(n) = \Theta(n \log n)$ in the best, average, and worst cases alike — Merge Sort's performance is remarkably stable and input-independent. Its main drawback is space: merging requires an auxiliary array, giving $O(n)$ extra memory, and every element is physically copied at every level of recursion (a total of $n \log n$ element copies), which contributes a non-trivial constant factor.

2.2 Quick Sort

Quick Sort is also divide-and-conquer, but it sorts *in place*. A pivot element is chosen, the array is partitioned so that elements smaller than the pivot precede it and larger elements follow it, and the two partitions are recursively sorted. If the pivot always splits the array evenly:

$$T(n) = 2T\left(\frac{n}{2}\right) + \Theta(n) \Rightarrow \Theta(n \log n) \quad (2)$$

However, if the pivot is chosen poorly (for example, always the last element of an already-sorted array), the partitions become maximally unbalanced:

$$T(n) = T(n-1) + \Theta(n) \Rightarrow \Theta(n^2) \quad (3)$$

Quick Sort's advantage is that it requires only $O(\log n)$ auxiliary space (for the recursion stack) and its inner loop touches memory sequentially, giving it excellent cache locality and a very small constant factor in the average case, at the cost of a fragile worst case.

2.3 Heap Sort

Heap Sort first builds a max-heap from the array in $O(n)$ time, then repeatedly removes the maximum element (the root) and restores the heap property by “sifting down,” an $O(\log n)$ operation performed n times:

$$T(n) = \underbrace{O(n)}_{\text{build heap}} + \underbrace{n \cdot O(\log n)}_{n \text{ extractions}} = O(n \log n) \quad (4)$$

Unlike Quick Sort, Heap Sort has *no* pathological worst case — it is $\Theta(n \log n)$ for every input. It is also in-place, needing only $O(1)$ extra memory. Its weakness is that the heap’s implicit tree structure means children of node i live at indices $2i + 1$ and $2i + 2$, which are not contiguous with i ; this causes more cache misses than Quick Sort’s sequential partitioning, inflating its constant factor.

2.4 Summary Table

Algorithm	Best	Average	Worst
Merge Sort	$n \log n$	$n \log n$	$n \log n$
Quick Sort	$n \log n$	$n \log n$	n^2
Heap Sort	$n \log n$	$n \log n$	$n \log n$

Table 1: Time complexity comparison. Merge Sort uses $O(n)$ extra space; Quick Sort and Heap Sort are in-place, using $O(\log n)$ and $O(1)$ extra space respectively.

3 Implementation

All three algorithms were implemented in C, instrumented with two global counters — `comparisons` and `swaps` — so the theoretical constants above could be checked empirically. Execution time was measured using the standard `clock()` function from `time.h`, which reports the number of CPU clock ticks consumed by the process; dividing by `CLOCKS_PER_SEC` converts this into seconds. This is the standard, portable way to measure “clock cycles” of a program in ANSI C, since architecture-specific cycle counters (e.g. `rdtsc`) are not portable.

3.1 Merge Sort Implementation

```

1 // Merges two sorted subarrays arr[left..mid] and arr[mid+1..right]
2 void merge(int arr[], int left, int mid, int right) {
3     int n1 = mid - left + 1;
4     int n2 = right - mid;
5     int *L = (int *)malloc(n1 * sizeof(int)); // O(n) extra space
6     int *R = (int *)malloc(n2 * sizeof(int));
7
8     for (int i = 0; i < n1; i++) L[i] = arr[left + i];
9     for (int j = 0; j < n2; j++) R[j] = arr[mid + 1 + j];
10
11     int i = 0, j = 0, k = left;
12     while (i < n1 && j < n2) {
13         comparisons++;
14         if (L[i] <= R[j]) arr[k++] = L[i++];
15         else arr[k++] = R[j++];
16     }
17     while (i < n1) arr[k++] = L[i++];
18     while (j < n2) arr[k++] = R[j++];
19     free(L); free(R);
20 }
21
22 // T(n) = 2T(n/2) + O(n) -> O(n log n) in ALL cases
23 void mergeSort(int arr[], int left, int right) {
24     if (left < right) {
25         int mid = left + (right - left) / 2;
26         mergeSort(arr, left, mid);
27         mergeSort(arr, mid + 1, right);
28         merge(arr, left, mid, right);

```

```

29     }
30 }

```

3.2 Quick Sort Implementation

```

1 // Lomuto partition: last element as pivot
2 int partition(int arr[], int low, int high) {
3     int pivot = arr[high];
4     int i = low - 1;
5     for (int j = low; j < high; j++) {
6         comparisons++;
7         if (arr[j] < pivot) {
8             i++;
9             swap(&arr[i], &arr[j]); // in-place: no extra array
10        }
11    }
12    swap(&arr[i + 1], &arr[high]);
13    return i + 1;
14 }
15
16 // Average:  $T(n)=2T(n/2)+O(n) \rightarrow O(n \log n)$ 
17 // Worst (sorted input):  $T(n)=T(n-1)+O(n) \rightarrow O(n^2)$ 
18 void quickSort(int arr[], int low, int high) {
19     if (low < high) {
20         int pi = partition(arr, low, high);
21         quickSort(arr, low, pi - 1);
22         quickSort(arr, pi + 1, high);
23     }
24 }

```

3.3 Heap Sort Implementation

```

1 // Sift down: restore max-heap property at index i
2 void heapify(int arr[], int n, int i) {
3     int largest = i, left = 2*i+1, right = 2*i+2;
4     if (left < n) { comparisons++; if (arr[left] > arr[largest]) largest = left; }
5     if (right < n) { comparisons++; if (arr[right] > arr[largest]) largest = right; }
6     if (largest != i) {
7         swap(&arr[i], &arr[largest]);
8         heapify(arr, n, largest);
9     }
10 }
11
12 // Build heap  $O(n)$  +  $n$  extractions  $O(\log n)$  each  $\rightarrow O(n \log n)$  ALWAYS
13 void heapSort(int arr[], int n) {
14     for (int i = n/2 - 1; i >= 0; i--) heapify(arr, n, i);
15     for (int i = n - 1; i > 0; i--) {
16         swap(&arr[0], &arr[i]);
17         heapify(arr, i, 0);
18     }
19 }

```

4 Experimental Setup

Each algorithm was run on *identical* random integer arrays (values in $[0, 99999]$) of increasing size, $n \in \{1000, 5000, 10000, 20000, 40000\}$ to ensure a fair comparison. Timing was captured immediately before and after each sort using `clock()`, and the difference in ticks (equal to `CLOCKS_PER_SEC = 1,000,000` on the test system) was converted to seconds. Comparison and swap counters were reset before each run.

5 Results

Algorithm	n	Clock Ticks	Time (s)	Comparisons	Swaps
Merge Sort	1,000	98	0.000098	8,730	0
Quick Sort	1,000	44	0.000044	10,821	5,291
Heap Sort	1,000	71	0.000071	16,896	9,098
Merge Sort	5,000	516	0.000516	55,228	0
Quick Sort	5,000	259	0.000259	72,695	38,077
Heap Sort	5,000	404	0.000404	107,674	57,149
Merge Sort	10,000	1,070	0.001070	120,497	0
Quick Sort	10,000	568	0.000568	153,612	82,119
Heap Sort	10,000	936	0.000936	235,350	124,263
Merge Sort	20,000	2,233	0.002233	260,914	0
Quick Sort	20,000	1,312	0.001312	323,938	171,674
Heap Sort	20,000	1,900	0.001900	510,945	268,522
Merge Sort	40,000	5,525	0.005525	561,734	0
Quick Sort	40,000	3,342	0.003342	739,787	386,584
Heap Sort	40,000	5,000	0.005000	1,101,654	576,926

Table 2: Empirical results across five input sizes. Clock ticks and time were measured with `clock()`; `CLOCKS_PER_SEC = 1,000,000` on the test system.

6 Discussion

6.1 Confirming $O(n \log n)$ scaling

For all three algorithms, doubling n roughly doubles the number of comparisons and multiplies the running time by a bit more than two, which is exactly the signature of $n \log n$ growth (as opposed to, say, quadratic growth, which would roughly quadruple). This is a direct empirical confirmation of the recurrences derived in Section 2.

6.2 Merge Sort: fewest comparisons, no swaps, but slower

Merge Sort consistently required the *fewest* comparisons of the three algorithms, since its merge step only ever compares the current heads of two already-sorted lists. It performed zero “swaps” in our instrumentation because it moves data through array copies rather than pairwise swaps. Despite this comparison efficiency, it was not the fastest algorithm in wall time, because every merge operation allocates auxiliary arrays and copies n elements at each of the $\log n$ levels of recursion — a cost that Big-O notation hides inside its constant factor.

6.3 Quick Sort: most swaps, fewest cycles

Quick Sort performed more comparisons than Merge Sort and roughly half as many swaps as Heap Sort, yet it was the fastest algorithm at every input size tested, using roughly 40–60% fewer clock ticks than Merge Sort and Heap Sort. This is explained by its in-place partitioning: elements are compared and swapped sequentially within a contiguous

block of memory, which is very friendly to CPU caching and requires no dynamic memory allocation, unlike Merge Sort’s per-call `malloc/free`. This illustrates the central lesson of the experiment: constant factors rooted in memory access patterns can matter more in practice than the raw comparison count.

It is worth stressing that this advantage is specific to *average-case, randomly ordered* input. Because this implementation uses the last element as pivot (Lomuto partitioning), an already-sorted or reverse-sorted array would trigger Quick Sort’s $O(n^2)$ worst case, and it would become far slower than either Merge Sort or Heap Sort. Production-quality implementations typically guard against this with randomized or median-of-three pivot selection.

6.4 Heap Sort: most comparisons and swaps, moderate speed

Heap Sort consistently required the most comparisons and swaps of the three algorithms. Each `heapify` call may compare a node against both children and then recurse, and the “sift-down” pattern touches indices $2i + 1$ and $2i + 2$, which are not adjacent to i in memory. This poor locality of reference explains why Heap Sort, despite having the same asymptotic bound as Merge Sort, ran at a similar or slightly faster wall-clock speed only because it avoids dynamic memory allocation, while still lagging noticeably behind Quick Sort throughout the experiment.

6.5 Constant factors versus asymptotic guarantees

The experiment highlights a practical trade-off:

- **Quick Sort** offers the best average-case constant factor but an unstable worst case, making it ideal when input is expected to be reasonably random or when a good pivot strategy is used.
- **Heap Sort** offers a guaranteed $O(n \log n)$ bound and $O(1)$ space with no risk of degradation, at the cost of a larger constant factor from poor cache locality — a good choice when worst-case guarantees matter (e.g. real-time systems).
- **Merge Sort** offers stability (equal elements retain their relative order) and a guaranteed $O(n \log n)$ bound, at the cost of $O(n)$ extra memory and the highest wall-clock time in this experiment — often preferred for linked lists or external sorting where sequential access dominates and stability is required.

7 Conclusion

This essay derived the theoretical recurrences governing Merge Sort, Quick Sort, and Heap Sort, implemented all three in C with instrumentation for comparisons and swaps, and measured their actual clock-cycle behaviour using `clock()`. The empirical results confirmed the expected $O(n \log n)$ scaling for all three algorithms, while also demonstrating that asymptotic complexity alone does not determine real-world performance: Quick Sort’s in-place, cache-friendly partitioning made it the fastest algorithm tested, despite not having the strongest worst-case guarantee, while Merge Sort’s auxiliary-array copying and Heap Sort’s non-contiguous memory access pattern both introduced measurable constant-factor overhead. This underlines an important lesson in algorithm design: Big-O notation is indispensable for reasoning about scalability, but engineering decisions about which algorithm to deploy in practice must also account for memory access patterns, space constraints, and the statistical properties of expected input.

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